

Taher Batterywala

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EDUCATION

VIT Bhopal University

B.Tech in CSE (Specialization in AIML)

Bhopal, India

Sep 2023 – Jun 2027

S.I.C.A. Senior Secondary School

Senior Secondary Education (CBSE, Class I-XII)

Indore, India

Apr 2008 – Mar 2023

TECHNICAL SKILLS

Programming Languages: C++, Python, Java, SQL, HTML/CSS, JavaScript

CS Fundamentals: Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks

AI & Machine Learning: Deep Learning, Computer Vision, Generative AI, Large Language Models (LLMs), NLP, Retrieval-Augmented Generation (RAG)

Frameworks & Tools: TensorFlow, Scikit-learn, LangChain, Pandas, Flask, Streamlit, Amazon Web Services (AWS), Git/GitHub, Render, REST APIs

EXPERIENCE

FOSSEE, IIT Bombay | Summer Intern | [GitHub](#)

May 2026 – Aug 2026

- Implemented a subprocess isolation layer utilizing Base64-serialized I/O pipes for a local AI Copilot, resolving C++ memory segmentation faults and eliminating UI thread latency (from 1.5s to near-zero).
- Integrated an offline RAG pipeline using Qwen2.5:3b and MiniCPM-V, tokenizing and indexing engineering manuals into 74 discrete ChromaDB vector database layers for local retrieval.
- Wrote a regex-driven garbage collection routine to eliminate file-system memory leaks from temporary clipboard images, and built a self-healing config script to resolve startup crashes on fresh installations.

PROJECTS

1. Tirakot AI: Offline Voice-Activated Desktop Assistant | [GitHub](#)

Local AI & Systems Engineering Project

Python, PyTorch, Ollama, Whisper, CustomTkinter, asyncio

- Engineered a dual-thread AI assistant running 3 ML models (Qwen 2.5, Whisper STT, Edge-TTS) concurrently under a strictly managed 2.34 GB VRAM budget on consumer hardware.
- Built 8 real-time OS automation tools (web search, app launching, VS Code generation) triggered by LLM JSON tool-calling via zero-latency first-token stream interception.
- Implemented an EMA-calibrated noise floor and 3-frame VAD gating for wake-word detection, completely eliminating Whisper transcription hallucinations during ambient silence.

2. GenWealth: Financial Forecasting & NLP System | [GitHub](#)

Deep Learning & Quant Finance Project

PyTorch (LSTM), Scikit-learn (Random Forest), FinBERT, yfinance

- Architected a hybrid forecasting engine (LSTM + Random Forest) trained on 200 equities across 16 years of market data, achieving 55% directional accuracy on unseen assets.
- Processed 1.84M raw headlines and executed GPU-accelerated FinBERT inference to generate daily confidence-weighted sentiment signals for 1.65M+ financial news records.
- Designed a deterministic "Logic Injection Layer" to correct model PermaBull bias, validated by a 7.2% confidence differential during live sentiment stress testing scenarios.

3. GamerSense: Behavioral Segmentation Engine | [GitHub](#)

Data Science & ML Engineering Project

Python, Scikit-learn, XGBoost, SMOTE, Flask, Pandas

- Engineered an end-to-end ML pipeline with dataclass configurations, deploying K-Means clustering to segment 3,024 mobile players into actionable behavioral personas.
- Benchmarked 7 algorithms utilizing SMOTE on a severe 38:6:1 imbalanced dataset, achieving 77.2% classification accuracy with an SVC to predict in-app spending segments.

LEADERSHIP & EXTRACURRICULARS

Google Developer Groups VIT Bhopal | Core Team Member | [Certificate](#)

Nov 2024 – Jun 2025

ACHIEVEMENTS & CERTIFICATIONS

- Competitive Programming** – Solved 300+ Data Structures and Algorithms problems on LeetCode using C++ [Profile](#)
- Introduction to Internet of Things (Topper 5%)** – NPTEL [Certificate](#)
- Applied Machine Learning in Python** – University of Michigan (Coursera) [Certificate](#)
- Cloud Computing** – NPTEL (Swayam) [Certificate](#)